FINM3013 Introduction to Financial Derivatives
Midterm Test

Name: _____ **Student ID:** _____

Date: 9 May 2025
Time allowed: 90 minutes **Full mark: 80**

I. Multiple choice questions (45 points):

*Identify the letter of the choice that **best** completes the statement or answers the question. 3 points for each question.*

1. The price of a stock on February 1 is $124. A trader sells 200 put options on the stock with a strike price of $120 when the option price is $5. The options are exercised when the stock price is $110. The trader's net profit or loss is
   A. Gain of $1,000
   B. Loss of $2,000
   C. Gain of $2,000
   D. Loss of $1,000

2. A short forward contract on an asset plus a long position in a European call option on the asset with a strike price equal to the forward price is equivalent to
   A. A short position in a call option
   B. A short position in a put option
   C. A long position in a put option
   D. None of the above

3. On March 1 a commodity's spot price is $60 and its August futures price is $59. On July 1 the spot price is $64 and the August futures price is $63.50. A company entered into futures contracts on March 1 to hedge its purchase of the commodity on July 1. It closed out its position on July 1. What is the effective price (after taking account of hedging) paid by the company?
   A. $59.50
   B. $60.50
   C. $61.50
   D. $63.50

4. You sell one December futures contract when the futures price is $1,010 per unit. Each contract is on 100 units and the initial margin per contract that you provide is $3,500. The maintenance margin per contract is $1,500. During the next day the futures price rises to $1,012 per unit. What is the balance of your margin account at the end of the day?
   A. $1,800
   B. $3,300
   C. $2,200
   D. $3,700

5. A company has a $36 million portfolio with a beta of 1.2. The futures price for a contract on an index is 900. Futures contracts on $250 times the index can be traded. What trade is necessary to increase beta to 1.8?
   A. Long 192 contracts
   B. Short 192 contracts
   C. Long 96 contracts
   D. Short 96 contracts

6. The yield curve is flat at 6% per annum. What is the value of an FRA where the holder receives interest at the rate of 8% per annum for a six-month period on a principal of $1,000 starting in two years? All rates are compounded semiannually.
   A. $9.12
   B. $9.02
   C. $8.88
   D. $8.63

7. The six month and one-year rates are 3% and 4% per annum with semiannual compounding. Which of the following is closest to the one-year par yield expressed with semiannual compounding? The par value is 100.
   A. 3.99%
   B. 3.98%
   C. 3.97%
   D. 3.96%

8. The six-month zero rate is 8% per annum with semiannual compounding. The price of a one-year bond that provides a coupon of 6% per annum semiannually is 97. What is the one-year continuously compounded zero rate? The par value is 100.
   A. 8.02%
   B. 8.52%
   C. 9.02%
   D. 9.52%

9. The term structure of interest rates is downward sloping. Define X as the five-year zero rate, Y as the yield on a five-year coupon-bearing bond and Z as the forward rate for the period between 4.75 and 5 years. Which of the following is true?
   A. X is less than Z which is less than Y
   B. Y is less than X which is less than Z
   C. Z is less than X which is less than Y
   D. Z is less than Y which is less than X

10. The spot price of an investment asset that provides no income is $30 and the risk-free rate for all maturities (with continuous compounding) is 10%. What is the three-year forward price?
    A. $40.50
    B. $22.22

C. $33.00
D. $33.16

11. A portfolio is worth $24,000,000. The quoted price for a Treasury bond futures contract is 110 and each contract is for the delivery of bonds with a face value of $100,000. On the delivery date the duration of the bond that is expected to be cheapest to deliver is 6 years and the duration of the portfolio will be 5.5 years. How many contracts are necessary for hedging the portfolio? Assume the face value of the Treasury bond is $100.
    A. 100
    B. 200
    C. 300
    D. 400

12. The time-to-maturity of a Eurodollars futures contract is 4 years and the time-to-maturity of the rate underlying the futures contract is 4.25 years. The standard deviation of the change in the short term interest rate, $\sigma = 0.011$. What is the difference between the futures and the forward interest rates?
    A. 0.105%
    B. 0.103%
    C. 0.098%
    D. 0.093%

13. Which of the following is true?
    A. The futures rates calculated from a Eurodollar futures quote are always less than the corresponding forward rate
    B. The futures rates calculated from a Eurodollar futures quote are always greater than the corresponding forward rate
    C. The futures rates calculated from a Eurodollar futures quote should equal the corresponding forward rate
    D. The futures rates calculated from a Eurodollar futures quote are sometimes greater than and sometimes less than the corresponding forward rate

14. What should a trader do when the one-year forward price of an asset is too low? Assume that the asset provides no income.
    A. The trader should borrow the price of the asset, buy one unit of the asset and enter into a short forward contract to sell the asset in one year.
    B. The trader should borrow the price of the asset, buy one unit of the asset and enter into a long forward contract to buy the asset in one year.
    C. The trader should short the asset, invest the proceeds of the short sale at the risk-free rate, enter into a short forward contract to sell the asset in one year
    D. The trader should short the asset, invest the proceeds of the short sale at the risk-free rate, enter into a long forward contract to buy the asset in one year

15. The modified duration of a bond portfolio worth $1 million is 5 years. By approximately how much does the value of the portfolio change if all yields increase by 5 basis points?
    A. Increase of $2,500
    B. Decrease of $2,500
    C. Increase of $25,000
    D. Decrease of $25,000

II. Problems (35 points)

1. (7 points) Suppose that the forward price for March 20 on euros is $0.9127 at the same time that the price of the euro futures for delivery on March 20 is $0.9145. How could an arbitrageur profit from this situation? What will be the arbitrageur's profit per futures contract (size is €125,000)?

2. (10 points) Assume that a bank can borrow or lend money at the same interest rate in the LIBOR market. The 90-day (i.e., 0.25 year) rate is 10% per annum, and the 180-day (i.e., 0.5 year) rate is 10.4% per annum, both expressed with continuous compounding. The Eurodollar futures price for a contract maturing in 90 days is quoted as 89.3. What arbitrage opportunities are open to the bank? Ignore the day count convention issue.

3. (18 points)

Suppose that the two-year interest rates in Australia and the United States are 5% and 7% per annum, respectively, with continuous compounding. The spot exchange rate is 0.6200 USD per AUD. In the following two cases, how could an arbitrageur make a profit?

a) (9 points) The two-year forward exchange rate is 0.6300.
b) (9 points) The two-year forward exchange rate is 0.6600.

Formula Sheet

- Conversion formulas between interest rates

$$R_c = m \ln\left(1 + \frac{R_m}{m}\right)$$
$$R_m = m\left(e^{R_c/m} - 1\right)$$

R_c is the interest rate with continuous compounding, and R_m is the rate with compounding m times per year.

- Par yield

$$c = \frac{(100 - 100d)m}{A}$$

where c is the coupon per annum, d is the present value of \$1 received at maturity, and A is the present value of an annuity of \$1 on each coupon data.

- Forward rate for the period between times T_1 and T_2

$$\frac{R_2 T_2 - R_1 T_1}{T_2 - T_1}$$

where R_1 and R_2 are zero rates for time periods T_1 and T_2 respectively, continuously compounded.

- Value of FRA

$$V_{FRA} = L(R_F - R_K)(T_2 - T_1)e^{-R_2 T_2}$$

Where R_K is the rate of interest agreed to in the FRA, R_F is the forward LIBOR interest rate for the period between T_1 and T_2, calculated today, R_2 is the continuously compounded riskless zero rate for maturity T_2, and L is the principal underlying the contract.

Table 5.6 Summary of results for a contract with time to maturity T on an investment asset with price S_0 when the risk-free interest rate for a T-year period is r.

Asset	*Forward/futures price*	*Value of long forward contract with delivery price K*
Provides no income:	$S_0 e^{rT}$	$S_0 - Ke^{-rT}$
Provides known income with present value I:	$(S_0 - I)e^{rT}$	$S_0 - I - Ke^{-rT}$
Provides known yield q:	$S_0 e^{(r-q)T}$	$S_0 e^{-qT} - Ke^{-rT}$

- Eurodollar futures contract value

10,000[100-0.25(100-Q)], where Q is the quoted price of a Eurodollar futures contract

- Relationship between forward and futures rates

Forward Rate = Futures Rate − $0.5\sigma^2 T_1 T_2$

- Duration

$$D = \sum_{i=1}^{n} t_i \left[\frac{c_i e^{-yt_i}}{B} \right]$$

- Duration-based hedge ratio

$$\frac{PD_P}{V_F D_F}$$